

Modern Algebra

Assignment 13

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13 Integral Domains

Problem 3

Proof. Suppose we have a commutative ring R with the cancellation property, that is, if $ba = bc$ then $a = c$ where $a, b, c \in R$. Suppose that a, b are non-zero elements. For the sake of contradiction, we assume that R has a zero-divisor b such that $ba = 0$. We know that $b \cdot a = b \cdot 0$. By the cancellation property, we cancel b , so $a = 0$, which contradicts our assumption that $a \neq 0$. Hence, a commutative ring cannot have a zero-divisor. ■

Problem 4

The set of zero-divisors of $\mathbb{Z}_{20}$, call it S , is: $\{2, 4, 5, 6, 8, 10, 12, 14, 15, 18, 16\}$.

We know that the set of units $U(\mathbb{Z}_{20}) = \{1, 3, 7, 9, 11, 13, 17, 19\}$

Thus, $S \cap U(\mathbb{Z}_{20}) = \emptyset$. The relationship between the two sets is that they are disjoint.

Problem 9

Consider $a = (2, 0, 1)$, $b = (3, 1, 0)$, $c = (0, 4, 1)$ and observe that $ab = (6, 0, 0)$, $bc = (0, 4, 0)$, $ac = (0, 0, 1)$. Clearly, ab, bc, ac are non-zero elements. In otherwords, none on them is the additive

identity of the ring $\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$. Observe that

$$ab \cdot bc = (6 \cdot 0, 0 \cdot 4, 0 \cdot 0) = (0, 0, 0)$$

$$bc \cdot ac = (0 \cdot 0, 4 \cdot 0, 0 \cdot 1) = (0, 0, 0)$$

$$a \cdot b \cdot c = (2 \cdot 3 \cdot 0, 0 \cdot 1 \cdot 4, 1 \cdot 0 \cdot 1) = (0, 0, 0).$$

Thus, $abc = (0, 0, 0)$ which is the additive identity, so it is not a zero-divisor. Hence, we found elements $a, b, c \in \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$ such that ab, bc, ac are zero divisors, and abc is not a zero divisor.

Problem 18

Proof. Suppose we have an integral domain D and a is an element in D . If a is idempotent, then $a^2 = a$. Observe that

$$a^2 = a$$

$$a^2 - a = a - a$$

$$a^2 - a = 0_D$$

$$a(a - 1) = 0_D$$

Then either $a - 1 = 0$ or $a = 0$, so $a = 1$ or $a = 0$. Hence, if there are idempotents in an integral domain then they are only 1 or 0. ■

Problem 32

Consider the ring $\mathbb{Z}_{10}$. Clearly, $R \subset \mathbb{Z}_{10}$. We need to show it is a subring. First, we show it is closed under subtraction, observe the following Cayley table:

−	0	2	4	6	8
0	0	8	6	4	2
2	2	0	8	6	4
4	4	2	0	8	6
6	6	4	2	0	8
8	8	6	4	2	0

Next, we show it is closed under multiplication, observe the following Cayley table: Hence,

·	0	2	4	6	8
0	0	0	0	0	0
2	0	4	8	2	6
4	0	8	6	4	2
6	0	2	4	6	8
8	0	6	2	8	4

R is a ring. From the previous table, we can see that if $a \in R$, then $a \cdot 6 = a$, so 6 is the

multiplicative identity. We see that 6 is in every row except the zero row, thus, every element has a multiplicative inverse. For commutativity, suppose that we have an index for each entry in the table ij , where $1 \leq i, j \leq 5$. From the multiplication table, the entries in the index ij are the same in the entries ji . Therefore, R is a field.

Problem 57

- a) We factor the expression to get $(x - 2)(x - 3) = 0$. We know that $\mathbb{Z}_7$ is an integral domain since 7 is prime number. Then, x is either 2 or 3. So we have two solutions for the equation $x^2 - 5x + 6 = 0$ in $\mathbb{Z}_7$.
- b) We factor the expression to get $(x - 2)(x - 3) = 0$. We know that $2, 3 \in \mathbb{Z}_8$, so $x = 2$ or $x = 3$ are solutions for the equation. $\mathbb{Z}_8$ is not an integral domain, then we can have other solutions for this equation other than $x = 2, x = 3$. However, there is no element $x \in \mathbb{Z}_8$ such that $(x \neq 2 \wedge x \neq 3) \wedge ((x - 2) \cdot (x - 3) = 0)$. Thus, the equation $x^2 - 5x + 6 = 0$ has only two solutions in $\mathbb{Z}_8$.